Research Proposal for the Ph.D. Thesis - Eiffel Scholarship

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Abstract

The aim of this Ph.D. research proposal is to use tools of applied mathematics to economic climate modeling. The first chapter develops continuous-time stochastic models with partial information, using filtering techniques in stochastic optimal control, where agents can probabilistically influence the states of the world—especially relevant for firms acting under uncertainty about climate damages and policies. The second chapter examines Mean Field Games (MFGs) under partial information, focusing on how limited knowledge of other agents' actions modifies the classical MFG framework, how the equilibrium differs from the perfect information case, and how to apply this approach to economic-climate problems. The third chapter addresses: (i) price dynamics driven by shock transmission in a network economy with price rigidities, modeled using a Graphon and MFG; and (ii) the interaction between economic growth, debt sustainability, and climate policies and damages, within a continuous-time stochastic framework.

Introduction

This research uses applied mathematics to improve economic models related to climate issues. Traditional models often miss important factors like uncertainty, how policies affect outcomes, and how different agents interact. By using methods such as stochastic optimization, Mean Field Games (MFG), and partial information, this study aims to create more accurate and useful climate-economic models.

In the first chapter, I develop a continuous-time dynamic stochastic model with partial information and a climate component. The elements of interest in this chapter are the approach to set up the model and solve a stochastic optimal control problem under a specific type of filtering problem induced by the partial information scheme on climate policies, and analyzing the solution of the stochastic optimal control problem where the agent can probabilistically influence the states (the policies) on which it is operating.

In the second chapter, I extend the core model to a Mean Field Game (MFG) within a partial information framework. This section studies an MFG model in a climate environment, where optimal climate policies of each agent, represented by abatement shares, depend on those of other agents. However, these policies are not directly observable. Consequently, the agent must solve a filtering problem. The focus is on differences in equilibrium outcomes under incomplete versus full information settings and their consequences on economic and climate performance.

The third chapter examines price dynamics in an economy under carbon taxes and the long-term dynamics of debt, economic growth, climate damage, and policies. The first model analyzes the effects of a carbon tax on prices within a network of interconnected firms, using an MFG model with heterogeneous interactions, represented by a *Graphon*. The second section explores the relationship between climate disasters, policies, economic growth, and debt sustainability, focusing on trade-offs in adopting climate policies.

This work is structured into three chapters, each addressing a specific research question within applied mathematics methods to economic-climate modeling. Each chapter is organized into three sections: one describing the research proposal and motivation, another outlining the draft mathematical framework, and the last explaining the future steps.

Chapter I

This chapter examines stochastic optimal control problems under partial information, focusing on economic-climate applications where agents can influence the state of the world. The model features a firm maximizing expected future profits while facing partially observed carbon tax policies, which it can probabilistically influence based on its behavior.

The research question and motivations

While stochastic optimal control with filtering is extensively studied, its application to climate models remains limited. This approach is motivated by the uncertainty that firms face regarding climate change impacts and government policies, which are typically overlooked in existing frameworks. Additionally, a firm's performance can influence future climate policies, allowing its actions to probabilistically affect policy scenarios. Specifically, the expected insights of this chapter are:

1. Analyze the impact of partial information (risk or ambiguity) on climate models, reflecting policy uncertainty and incomplete climate understanding.

2. Investigate stochastic optimal control problems where agents can probabilistically alter the state of the world under partial information of such states, considering how firm actions influence climate policies like carbon taxes.

3. Model partial information in climate-economic problems and apply filtering solutions to derive policy insights.

The model to be developed in this chapter builds on frameworks such as [20] and [16], extending recent efforts to integrate climate change into continuous-time models (e.g., [11]). Recent work addresses *Knightian uncertainty* using decision-theoretic tools like ambiguity aversion, as seen in [17]. A similar work is present in [14], but this work incorporates filtering problems and explicitly links firms' emissions to the process for emission taxes. This chapter contributes to the literature by enhancing the understanding of stochastic optimal control with partial information in climate models, an underexplored area. It addresses the typically neglected uncertainty in climate events and policies and elucidates the dynamic interaction between firm behavior and environmental regulations, offering a robust framework for strategic decision-making under uncertainty.

Mathematical Framework

An agent aims to maximize expected discounted profits, which are negatively affected by an emissions tax on production-related activities. The emissions tax is modeled using a stochastic differential equation that includes both drift and volatility. The firm's optimization problem is given by

$$\sup_{(I_t,E_t)} \mathbb{E}\left[\int_0^\infty e^{-\rho t}\,\Pi_t\big(P_t^y, Y_t, C\big(Y_t, P_t^\tau\big)\big)\,dt\right],$$

where P_t^y is the price of output Y_t, and $C(Y_t, P_t^\tau)$ is a cost function depending on output and the emissions tax P_t^τ. The above problem is subject to a specific process for the variables in the payoff functional. Clearly, the firm must compute an expectation of the variables on which the profit function depends. In the case of carbon taxes, the process is given by

$$dP_t^\tau = \alpha_t^i\,dt + \sigma_\tau\,dW_t^\tau.$$

I assume that the drift $\alpha_t^i \in \{\alpha_L, \alpha_H\}$ is a stochastic process depending on the emissions E_t and is not fully observed by the firm. Thus, the firm must form an optimal belief about this variable to compute expectations. This estimation can be addressed using filtering techniques like the Wonham filter, which provides an optimal belief of the unobservable variable based on observable variables. Additionally, stochastic changes in α_t^i depend on the agent's actions (for example, how much to produce, leading to emissions E_t that modify α_t^i). Therefore, the agent must account for how its actions influence policies, represented by α_t^i, while relying on partial information.

Future developments

The aim of this chapter is to explore the above model to answer the research questions. To achieve this, the model should be further developed to ensure greater consistency and coherence. Moreover, the stochastic optimal control problem and the filtering approach must both be formulated in a mathematically rigorous manner. Once this has been accomplished, it will be possible to obtain initial results, either by deriving analytical solutions or conducting numerical simulations, and to analyze the effects of partial information on climate models as well as the characteristics of a stochastic optimal control problem where agent controls can modify the states of the world. Additionally, different schemes involving partial information or control-state relationships can be further developed.

Chapter II

This chapter aims to develop a Mean Field Game model with partial information tailored to climate modeling, incorporating heterogeneous abatement costs among multiple agents. It seeks to analyze how partial information influences equilibrium outcomes compared to full information scenarios.

The research question and motivations

Although some theoretical results have been obtained in the study of Mean Field Games (MFG) with partial information, very few results have been achieved regarding their application to practical problems in economics, finance, and climate.

The aim of this chapter is to develop a draft of an MFG model with partial information, focusing on climate modeling, while allowing for extensions to similar problems in economics and finance. Specifically, the objective

is to establish a framework that enables exploration of the practical differences in equilibria between MFG models with and without partial information, and to examine how the theory of stochastic optimal control under uncertainty can be applied to this context. The expected insights of this chapter are:

1. How to address a partial-information environment in MFGs, specifically in those dealing with climate damage and policy environments? What are the general properties of MFG modeling under partial information?

2. How does the equilibrium of such an MFG depend on the partial-information scheme? Specifically, how does this equilibrium differ from a full-information scenario?

3. Under a partial-information scheme, what do optimal policies to reduce emissions look like while minimizing welfare loss?

There is extensive literature on the relationship between stochastic optimal control and partial information. However, as mentioned in [7], the theoretical and applied contributions to Mean Field Games (MFG) with partial information are limited. Some of these contributions include [7], [8], and [12]. In this chapter, the aim is to contribute to the literature by creating an MFG that introduces partial information specifically to include climate modeling. Unlike most existing MFG studies that assume full information, this work addresses the realistic scenario where agents operate under uncertainty and information asymmetry, which is crucial for complex domains like climate policy.

Mathematical Framework

Consider infinitely many agents normalized to one, each maximizing expected discounted consumption. Production emits pollutants, reducing output via a damage function D. Abatement μ_t^i lowers emissions but has a cost $\xi_t(\mu_t^i, \mu_t^M)$, where μ_t^M is the median of the abatement distribution $\overline{\mu}_t$. The agent's problem is

$$\sup_{(C_t^i, \mu_t^i, I_t)} \mathbb{E}\left[\int_0^\infty e^{-\rho t} U\big(C_t^i - \xi(\mu_t^i, \mu_t^M)\big)\, dt\right],$$

subject to dynamics for the state variables, and specifically for the emissions, given by

$$dE_t^i = (1 - \mu_t^i) E_t^i\, dt + \sigma^i E_t^i\, dW_t^{E_i}.$$

Under partial information, the agent cannot observe $\overline{\mu}_t$. Instead, the agent infer it from noisy emissions from E_t, treating $\overline{\mu}_t$ as a hidden (distribution) state. An optimal belief π_t of the hidden state can be obtained using the Zakai equation:

$$d\widetilde{\pi}_t^i(x) = \mathcal{L}[\widetilde{\pi}_t^i](x)\, dt + \widetilde{\pi}_t^i(x)\, H(x)\, dZ_t^i,$$

where H and Z_t^i are part of the observable equations. The agent updates an unnormalized belief $\widetilde{\pi}_t^i$ that evolves under filtering dynamics of the above PSDE, and then obtains the normalized posterior π_t^i. This filtered belief enters the agent's optimization, leading to an MFG equilibrium that differs from the perfect-information case.

Future developments

Building on the draft structure, this chapter aims to define the problem and the MFG system. Once the problem and the MFG system are well-defined, the focus will be on establishing existence and uniqueness of solutions and identifying analytical forms to understand agent behavior under partial information. Numerical methods, possibly requiring specialized tools for the filtering framework, will help compute solutions and compare models with and without partial information. Finally, the chapter will derive policy implications from the analysis.

Chapter III

The aim of this chapter is to establish two mathematical models: one that enables understanding of price dynamics under carbon taxes within a network, and another that aims to understand the relationship among economic growth, fiscal policies, and climate change.

The research question and motivations

Price dynamics under carbon tax The first model aims to understand how carbon taxes affect prices within a networked economy. While the impact of carbon taxes on prices has been widely studied [23, 13], this chapter introduces an interconnected component among different economic sectors using Graphons. Unlike representative-agent models, this study employs Mean Field Games (MFG) to account for heterogeneity and agent interactions, which are often neglected in current economic models. The expected insights include:

1. Comparing shock transmission under a network structure versus a non-network setting, understanding how connections among sectors transmit shocks.

2. Exploring the use of applied mathematics tools and Graphons in MFGs and their applications in economics and finance, specifically regarding policy insights on how carbon taxes should be adopted.

3. Include time-varying or stochastic Graphons to capture evolving sector relationships.

Traditional economic network models rely on discrete-time frameworks with finite input-output matrices [18, 5, 22]. Recent advances combine price frictions with MFGs [2, 3, 21]. This research addresses gaps by integrating a network approach with MFGs and introducing Graphons to model network structures, a novel approach in price rigidity models.

Economic Growth, Fiscal Performance, and Climate The second model studies how economic growth, debt, and climate policies interact. While most literature addresses short-term carbon tax effects [25, 10, 6, 9, 24], a gap remains on how growth and debt respond to climate change. The expected insights from this chapter are:

1. How to model a stochastic optimal control problem to analyze long-term relationships between growth, debt sustainability, climate damages, and policies?

2. What are the optimal climate policy interventions, and how do stochastic climate shocks affect fiscal stability and debt dynamics over time?

3. How do different climate policy trajectories influence the sustainability of public debt over the long term, identifying policy paths that avert fiscal crises?

While studies on growth and climate change are extensive [19, 1, 4, 15], few address debt, growth, and climate within a unified framework. This model examines long-term outcomes of climate action or inaction, highlighting how policies and disasters can affect growth, fiscal sustainability, and the risk of default or zero-growth scenarios.

Mathematical Framework

Carbon tax and price dynamics Assume a unit mass of sectors; in each sector there is a firm $s \in [0,1]$. Each firm s seeks to maximize expected discounted profits Π, depending on the units produced and sold, the cost function, the tax on emissions generated when producing, and the abatement costs. The production function is

$$Y_t^s = A^s \left(\int_0^1 \Theta\big(\mathcal{G}_t(s,-s), Y_t^{-s}\big)\, ds \right),$$

where Θ is a mapping that defines how inputs are combined to generate output in that sector. To model network relationships, I use a Graphon $\mathcal{G}(s,-s)$. This function represents the network structure across sectors (each sector uses output from other sectors to produce). When producing output, firm s generates emissions $e_{s,t}$, which can be modeled as a function of output: $e_{s,t} = \phi\big(Y_t^s\big)$.

Next, I assume that there are price frictions in this economy. Specifically, firms cannot freely adjust their prices. Each time they wish to do so, they must pay an amount defined by the function ξ. The profit problem then becomes

$$\Pi_t^s = \text{Demand} \cdot P_{s,t} \;-\; \text{Cost} \;-\; \tau_{s,t} \cdot e_{s,t} \;-\; \xi_t\big(dP_{s,t}\big).$$

The price $P_{s,t}$ that maximizes the profit function is denoted by $P_{s,t}^*$. I define the log gap between the actual price of firm s and its optimal price as $m_{s,t} = \ln\big(P_{s,t}\big) \;-\; \ln\big(P_{s,t}^*\big)$. With this definition, the problem can be rewritten as choosing the optimal updates of the gap $m_{s,t}$, and the overall model is set up as an MFG, where the firm will look at the distribution of price gaps when choosing the optimal updates of its own price (since the optimal price gap updates will depends on such distribution, usually via the demand of the firm output).

Debt, economic growth, and climate For the second model, I have developed a basic framework to understand the relationships between climate change and policies, economic growth, and fiscal conditions. I focus on the trade-offs between current consumption, mitigation policies, debt sustainability, and growth. The problem to be solved is

$$\sup_{C_t, I_t, b_t, G_t, \mu_t} \mathbb{E}\left[\int_0^\tau \xi_t\, U\big(C_t \;-\; \text{Cost}_\mu \;-\; D^c(G_t, E_t)\big)\, dt \right],$$

where $\tau = \inf\Big\{ t \in [0,T] : \frac{B_t}{Y_t} \geq \overline{\theta} \in \mathbb{R}_+ \Big\}$ is a stopping time, B_t is the accumulated debt, Y_t is output, C_t represents consumption, D^c and D^y are damage functions on consumption and production dependent on emissions E_t and public spending G_t, and Cost_μ denotes the abatement cost μ_t. Public spending G_t helps mitigate climate shocks through transfers. Lastly, ξ_t is a generic discounting function that allows for *present bias*. The payoff functions are coupled with equations describing the dynamics of the variables.

The aim of the above model is to study the trade-off where fewer green policies today may worsen future fiscal conditions, potentially reaching a breaking point. A low-abatement and high-debt policy can boost short-term production and consumption but may lead to higher emissions, increased damages, and rising debt, ultimately risking default.

Future developments

The price dynamics model will be formulated as a Mean Field Game (MFG) with a Graphon structure to capture network connections among sectors. This framework analyzes how carbon taxes influence prices through sector interactions. Numerical methods will simulate price dynamics and compare outcomes with and without network linkages.

The second model, focused on debt, growth, and climate, will be refined to better capture long-term relationships. It examines how delaying climate investments and increasing debt heighten fiscal and environmental risks, potentially causing economic crises or zero growth. The model also considers poor climate governance, where climate shocks destabilize fiscal systems and increase default risks. Debt dynamics and feedback loops will be included, modeling how climate shocks affect fiscal stability. Present bias will be incorporated to reflect underestimation of future climate impacts. Ensuring existence and uniqueness of solutions will be key to robustness.

Finally, both models will be integrated to link short-term price dynamics with long-term economic and fiscal outcomes. Analytical and numerical methods will explore scenarios, filling gaps in the literature by analyzing price dynamics, growth, debt, and climate policies in a networked setting.

References

[1] Daron Acemoglu, Philippe Aghion, Leonardo Bursztyn, and David Hemous. The environment and directed technical change. *American Economic Review*, 102(1):131–66, February 2012.

[2] Yves Achdou, Jiequn Han, Jean-Michel Lasry, Pierre-Louis Lions, and Benjamin Moll. Income and wealth distribution in macroeconomics: A continuous-time approach. *The Review of Economic Studies*, 89(1):45–86, 04 2021.

[3] Fernando Alvarez, Francesco Lippi, and Panagiotis Souganidis. Price Setting With Strategic Complementarities as a Mean Field Game. *Econometrica*, 91(6):2005–2039, November 2023.

[4] Mohd Arshad Ansari, Vaseem Akram, and Salman Haider. A link between productivity, globalisation and carbon emissions: evidence from emissions by coal, oil and gas. *Environmental Science and Pollution Research*, 29(22):33826–33843, 2022.

[5] David Rezza Baqaee and Emmanuel Farhi. Macroeconomics with Heterogeneous Agents and Input-Output Networks. NBER Working Papers 24684, National Bureau of Economic Research, Inc, June 2018.

[6] Lint Barrage. The fiscal costs of climate change. *AEA Papers and Proceedings*, 110:107–12, May 2020.

[7] Alain Bensoussan and Sheung Chi Phillip Yam. Mean field approach to stochastic control with partial information, 2019.

[8] Charles Bertucci. Mean field games with incomplete information, 2023.

[9] Hannes Boehm. Physical climate change and the sovereign risk of emerging economies. *Journal of Economic Structures*, 11(1):31, 2022.

[10] Patrick Bolton, Lee Buchheit, Mitu Gulati, Ugo Panizza, Beatrice Weder di Mauro, and Jeromin Zettelmeyer. On debt and climate. *Oxford Open Economics*, 2:odad005, 07 2023.

[11] Rene Carmona, Gokce Dayanikli, and Mathieu Lauriere. Mean field models to regulate carbon emissions in electricity production, 2021.

[12] Philippe Casgrain and Sebastian Jaimungal. Mean field games with partial information for algorithmic trading, 2019.

[13] Günter Coenen, Romanos Priftis, and Matija Lozej. Macroeconomic effects of carbon transition policies: An assessment based on the ecb's new area-wide model with a disaggregated energy sector. ECB Working Paper 2023/2819, European Central Bank, May 2023. Available at SSRN.

[14] Katia Colaneri, Rüdiger Frey, and Verena Köck. Random carbon tax policy and investment into emission abatement technologies, 2024.

[15] Samuel Fankhauser and Richard S.J. Tol. On climate change and economic growth. *Resource and Energy Economics*, 27(1):1–17, 2005.

[16] Wendell H. Fleming and Tao Pang. A stochastic control model of investment, production and consumption. *Quarterly of Applied Mathematics*, 63(1):71–87, 2005.

[17] Andrea Mazzon and Peter Tankov. Optimal stopping and divestment timing under scenario ambiguity and learning, 2024.

[18] Ronald E. Miller and Peter D. Blair. *Input-Output Analysis: Foundations and Extensions*. Cambridge University Press, 3 edition, 2022.

[19] William D. Nordhaus. Economic growth and climate: The carbon dioxide problem. *The American Economic Review*, 67(1):341–346, 1977.

[20] Huyên Pham. Continuous-time stochastic control and optimization with financial applications / huyen pham. 2009.

[21] Valerio Pieroni. Energy shortages and aggregate demand: Output loss and unequal burden from hank. *European Economic Review*, 154:104428, 2023.

[22] Elisa Rubbo. Networks, Phillips Curves, and Monetary Policy. *Econometrica*, 91(4):1417–1455, July 2023.

[23] Jean-Guillaume Sahuc, Frank Smets, and Gauthier Vermandel. The new keynesian climate model. November 26 2024. Available at SSRN.

[24] Caterina Seghini. Sovereign debt sustainability, the carbon budget and climate damages. Research Paper No. 24-15, Swiss Finance Institute, February 2024. Available at SSRN: https://ssrn.com/abstract=4723223 or http://dx.doi.org/10.2139/ssrn.4723223.

[25] Stavros A. Zenios. The risks from climate change to sovereign debt. *Climatic Change*, 172(3):30, 2022.